PAIN® 154 (2013) 2388–2396

PAIN®

www.elsevier.com/locate/pain

Effect of various kinds of cervical spinal surgery on clinical outcomes: A systematic review and meta-analysis

Arianne P. Verhagen [a,*], Marienke van Middelkoop [a], Sidney M. Rubinstein [b], Raymond Ostelo [b,c], Wilco Jacobs [d], Wilco Peul [d,e], Bart W. Koes [a], Maurits W. van Tulder [b,c]

[a] Department of General Practice, Erasmus MC, University Medical Center Rotterdam, The Netherlands
[b] Department of Epidemiology and Biostatistics, EMGO-Institute for Health and Care Research, VU University Medical Center, Amsterdam, The Netherlands
[c] Department of Health Sciences, Faculty of Earth and Life Science and EMGO-Institute for Health and Care Research, VU University, Amsterdam, The Netherlands
[d] Department of Neurosurgery, Leiden University Medical Center, Leiden, The Netherlands
[e] Medical Center Haaglanden, The Hague, The Netherlands

Sponsorships or competing interests that may be relevant to content are disclosed at the end of this article.

ARTICLE INFO

Article history:
Received 3 May 2013
Received in revised form 12 July 2013
Accepted 15 July 2013

Keywords:
Systematic review
Meta-analysis
Cervical disorder
Surgery

ABSTRACT

The choice of a specific surgical technique should be based on its benefits and harms. Previous reviews have shown that the benefit of surgery over conservative care is not clearly demonstrated in patients with disorders of the cervical spine. Also, no additional benefit of fusion upon anterior decompression techniques could be found. A clear overview of other surgical techniques is lacking. We therefore aimed to assess the benefits and harms of cervical spinal surgery in patients with cervical disorders.

We searched MEDLINE, EMBASE, CINAHL, and CENTRAL up to June 2012. Randomized controlled trials (RCTs) were selected which included adults with cervical disorders receiving a surgical intervention and that reported at least 1 clinically relevant outcome measure (eg, pain, function, recovery). Two authors independently assessed the risk of bias using the criteria recommended by the Cochrane Back Review Group and extracted the data. The quality of the evidence was rated using the GRADE method. We included 39 RCTs comparing different surgical interventions. We found low-quality evidence for no difference in effectiveness between various surgical techniques used for anterior discectomy. There is a small, clinically irrelevant benefit on recovery and pain in favour of prosthetic disc surgery when compared with fusion techniques. Unfortunately, in these studies the authors had a clear conflict of interest. The differences in benefits and harms between the various surgical techniques are small. The surgeon, patient, and health care provider can therefore make the choice of any surgical technique based on experience, preferences, or costs.

© 2013 International Association for the Study of Pain. Published by Elsevier B.V. All rights reserved.

1. Introduction

Cervical disorders are common diseases that are frequently disabling and costly [13,32,35,50,64]. Patients with cervical disorders can be classified into a small group with specific (eg, tumours, fractures) and degenerative (cervical degenerative disorder, cervical radiculopathy and cervical myelopathy) disorders and a larger group of patients with nonspecific complaints. Patients with cervical disorders are initially given conservative therapy. When patients do not respond to conservative treatment they can be referred to a neurosurgeon, who may consider a surgical intervention. Patients with a cervical degenerative disorder often have chronic neck pain without radiculopathy; the goal of surgery is to reduce pain and improve function. Patients with cervical radiculopathy usually suffer from acute pain; the goal of surgery is to improve arm pain in the short term and avoid further neurological complications.

The most effective type of surgical procedure for cervical disorders is unknown. Anterior cervical discectomy (ACD) is a commonly used procedure and aims to relieve pressure on the nerve roots or the spinal cord by removing the ruptured disc. ACD is frequently combined with fusion (ACDF). The structural support for fusion is provided by different materials, including bone autografts (implant material derived from the same individual) or allografts (implant material from another individual), cages, plates, implants, and prostheses. In the last decade ceramic,

* Corresponding author. Address: Department of General Practice, Erasmus MC, University Medical Center Rotterdam, PO Box 2040, 3000 CA Rotterdam, The Netherlands. Tel.: +31 10 7044109.
E-mail address: a.verhagen@erasmusmc.nl (A.P. Verhagen).

0304-3959/$36.00 © 2013 International Association for the Study of Pain. Published by Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.pain.2013.07.022

carbon, and polyetheretherketone (PEEK) cages and cervical artificial discs have been developed.

Several systematic reviews have focused on the effectiveness of ACD, ACDF and prosthetic discs [4,17,26,30,39,40]. A recent Cochrane review [26] has updated an earlier review [30] and includes 33 trials. It concluded that radiological outcomes show that autograft fusion gives a better chance for fusion than a cage but found no differences in clinical outcomes. Two other reviews found, respectively, no clear clinical benefit of surgery over conservative care [39] and no additional benefit of fusion upon ACD [40]. Two recent systematic reviews evaluated ACDF in comparison with prosthetic disc surgery or arthroplasty [4,17], both including 9 trials (with 4 trials in common), and neither found a clear clinical benefit for either intervention.

New trials have recently been published. Furthermore, the effectiveness of artificial discs is still unclear, and most reviews do not focus on patient-reported outcomes and harms. There exists no overall systematic review of the benefits and harms of cervical spinal surgery. The objective of this study was therefore to evaluate the benefits and harms of surgical techniques in comparison with each other in patients with cervical disorders based on patient-reported outcomes such as pain, function, recovery, and reoperation.

2. Methods

2.1. Data sources and search strategy

We followed the search strategy outlined by the Cochrane Back Review Group (CBRG). The primary search was conducted in MEDLINE, EMBASE, CINAHL, and CENTRAL (the Cochrane Central Register of Controlled Trials) up to June 2012. An experienced librarian developed the search strategy by using free text words and MeSH headings. Next we screened references from the included studies and contacted experts in the field in order to identify additional studies. The language was limited to English, Dutch, and German.

Two review authors (SMR and MvM) independently selected citations based on titles and abstracts. They then assessed the eligibility of the full papers of those citations thought to fulfil the inclusion criteria. A third review author (APV) was consulted if consensus was not reached.

2.2. Study selection

Only randomized controlled trials (RCT) including adult ($\geqslant$18 years of age) subjects with cervical disorders were considered. We excluded patients with distinct cervical myelopathy who needed specific surgery, as this is a very serious condition. We included studies evaluating anterior cervical surgical interventions, including discectomy with different interbody fusion methods and prosthetic disc surgery. Primary outcome measures of interest were: neck and/or arm pain, neck-pain-specific functional status, and self-perceived recovery. Secondary outcomes were sick leave and complications of surgery. Studies were combined into 2 clusters:

(1) Surgical fusion methods compared with other surgical fusion methods.
(2) Surgical fusion methods compared with artificial (prosthetic) cervical disc surgery.

2.3. Data extraction

We used a standardized form to extract data from the included papers: study characteristics, patient characteristics, description of the experimental and control interventions, cointerventions (if used), duration of follow-up, types of outcomes assessed, and possible conflict of interest. Two review authors (SMR and MvM) independently extracted data and disagreements were discussed. A third review author (APV) was consulted if necessary.

2.4. Risk of bias

Two review authors (SMR and MvM) independently evaluated the risk of bias (RoB) using the 12 criteria recommended by the CBRG [20]. The criteria were scored as "yes", "no" or "unclear". Any disagreements between the review authors were resolved by discussion, including input from a third review author (APV). We defined a study with a low RoB when it fulfilled 6 or more of the criteria items and did not suffer from serious flaws as supported by empirical evidence [61].

2.5. Data synthesis and analysis

We decided to pool the data when studies were clinically homogenous (according to the experts); point estimates and measures of variation are reported for continuous outcomes and numbers of events for dichotomous outcomes. We used Review Manager 5.1 for the analysis (random effects model). The follow-up time intervals were defined as: short term (1–12 months) and long term (12 months or more). In cases of multiple follow-up measurements or follow-up measurements of only >3 months and <24 months we chose the closest interval.

We calculated mean differences (MD) for continuous outcomes. All visual analogue scales (VAS) were converted to scales ranging from 0–100 where necessary. For dichotomous outcomes we calculated risk differences (RD). The event was defined as the number of subjects not recovered; dropouts were regarded as not recovered. The clinical relevance of the results was set at a difference of $\geqslant$15% between groups. When possible we evaluated publication bias using a funnel plot and performed sensitivity analyses based on RoB and conflict of interest.

We used GRADE to assess the overall quality of the evidence and strength of recommendations [22]. The quality of the evidence for a specific outcome was downgraded by one level for each of the following factors that were encountered:

- Limitations in design (>25% of the participants from studies with a high RoB).
- Inconsistency of results (significant statistical heterogeneity [I^2 >40%] or inconsistent findings among studies [$\leqslant$75% of the participants reported findings in the same direction]).
- Indirectness (ie, generalizability of the findings).
- Imprecision (total number of participants <300 for a dichotomous outcome and <400 for continuous outcome).
- Other (eg, publication bias, flawed design, massive dropout).

The following levels for the quality of the evidence were applied:

- *High quality:* Further research is very unlikely to change the level of evidence; all domains are fulfilled.
- *Moderate quality:* Further research is likely to have an important impact on confidence in the estimate of effect and may change the estimate; 1 of the domains is not fulfilled.
- *Low quality:* Further research is very likely to have an important impact on confidence in the estimate of effect and is likely to change it; 2 of the domains are not fulfilled.
- *Very low quality:* Great uncertainty about the estimate; 3 of the domains are not fulfilled.

Single randomized studies (n <300 or <400) were considered inconsistent and imprecise and provided low-quality evidence, which could be further downgraded to very low-quality evidence.

3. Results

3.1. Study selection

We found 52 publications of 39 RCTs comparing different surgical techniques (Fig. 1). Several studies had been published more than once. After personal communications with the authors, 8 publications were considered to represent the same single large-scale multicenter trial [1,11,24,25,52–54,58]. One study had been published 4 times [2,15,29,42], one 3 times [67–69], and two other studies twice [5,41,7,8]; the latter presenting more patients in the 3-year follow-up publication than the former. See Supplementary Tables 1 and 2 for the study characteristics.

3.2. Description of studies

3.2.1. Participants

The total number of participants in the studies was 4212. Most studies included patients with neck pain and cervical radiculopathy or degenerative cervical disc disease, often confirmed by MRI and refractory to conservative treatment. This means these studies included patients with long-term complaints. The content and duration of conservative treatment remained unclear in all studies. Five studies did not describe the selection of patients very clearly [6,36,56,57].

3.2.2. Interventions

A wide variety of fusion techniques were used, of which the "classical" iliac crest autograft and PEEK cage fusion were the most frequent. Twenty-nine studies compared various fusion techniques [3,6,9,10,14,18,19,21,23,28,31,33,34,36–38,44,46–48,51,55–7,59,62,63,65,67]. Three studies compared 3 fusion techniques [10,23,65] and one study compared 6 different fusion techniques [38]. The mean sample size was 75 patients (range 24–267), 14 studies (48.2%) had study arms with <25 patients included.

Ten studies compared fusion vs prosthetic disc surgery [8,12,25,27,41–43,45,49,66]. The mean sample size was 203 patients (range 20–582) and 2 studies (22.2%) had a study arm with <25 patients included. The prosthetic discs varied between the Bryan ($n = 3$), Prestige ($n = 2$), Kineflex-C ($n = 2$), ProDisc-C ($n = 2$), and one other.

3.2.3. Outcome measures

Main outcome measures were pain (VAS), function using the neck disability index (NDI), or recovery, often using Odom's criteria (a 4-point scale ranging from excellent to poor; which is dichotomised between recovered and not recovered). Most studies had a follow-up period of 24 months, for three studies it was 6 months and six studies had a follow-up period ranging from 12 to 52 months [2,9,10,47,56,63].

Adverse events were reported in 28 of the 38 studies; the most frequent were donor-site pain and reoperation either due to lack of fusion or of an adjacent level.

Ten studies (25.6%) did not present any point estimates or measures of variability and could therefore not contribute to the statistical pooling.

3.2.4. Conflict of interest

In 10 studies there was a clear statement of a conflict of interest. Six other studies mentioned funding from a funding body but it was unclear whether there was a potential conflict of interest. In 11 studies conflict of interest was unknown and another 12 studies stated there was no conflict of interest, but in 1 study two of the authors were stockholders of the company responsible for the intervention [63].

3.3. Risk of bias

Only 2 RCTs had a low RoB (see Supplementary Table 3) [18,42]. The most common item that scored positive was compliance (100%), mainly because compliance with surgery is often

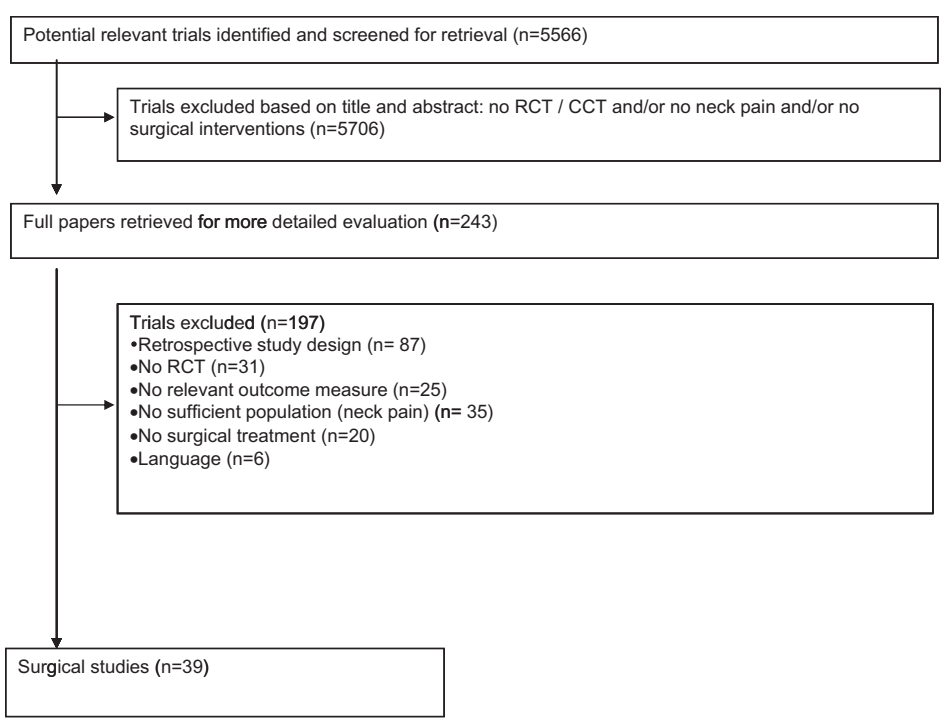

Fig. 1. Flow chart.

acceptable when included in a trial evaluating 2 or more surgical interventions.

Items that were unclear or scored negative in all studies were blinding of the patients, the care provider, and the outcome assessor; and the intention-to-treat analysis applied. Most of the studies scored unclear on concealment of allocation and cointerventions avoided or similar. The item on selective outcome assessment was scored unclear because there were no study designs published in trial registries. Overall, the studies comparing fusion with prosthetic disc surgery scored more items positive (37.5%) than the fusion vs fusion studies (27.3%).

3.4. Effectiveness

3.4.1. Surgical fusion methods compared with other surgical fusion methods

Twenty-nine studies evaluated a fusion technique in comparison with another fusion technique, of which one had a low RoB [18]. The results are presented so that the more experimental fusion techniques (fusion 2) are compared with the standard fusion technique (fusion 1). The standard technique was iliac crest autograft but PEEK cervical interbody cages replaced this over time. Both techniques were regarded as standard.

3.4.1.1. Recovery. Seventeen studies measured recovery at long-term follow-up using Odoms' criteria, 3 studies used a Japanese Orthopaedic Association score [9,14,38], 2 studies used the Prolo scale [10,18], and one study used a 5-point recovery scale [33]. Two studies additionally presented a short term (6-month) outcome [19,28]. Overall, 78.2% of patients recovered. A funnel plot showed possible publication bias (data not presented). Two studies did not provide data on recovery, 2 others presented a continuous outcome, and therefore 13 studies contributed to the statistical pooling (Fig. 2). Studies were ranked according to year of publication. Studies were statistically homogeneous ($I^2 = 0\%$) and no trends over time could be found in the forest plot (Fig. 2); we therefore considered pooling the results to be legitimate. The pooled effect estimate (13 studies; n = 880) showed a nonsignificant RD of 5% (−0% to 9%) in favour of the newer technique. When selecting only the studies with no conflict of interest (n = 278) [19,23,28,62] the RD was 5%, but was less precise (−2% to 13%). When selecting studies that stated the inclusion of patients with cervical radiculopathy (n = 8; see Supplementary Table 1) we found an RD of 6% (−0% to 13%).

3.4.1.2. Pain. Nineteen studies measured pain (frequently, neck and arm pain separately); 17 studies used a VAS. Only 7 studies adequately reported data, of which 2 reported dichotomous data [33,66], 4 reported continuous data on neck and arm pain separately [6,44,46,59] and 2 reported continuous data on overall pain [51,59].

We found no significant differences between the groups with respect to short- and long-term neck pain (Fig. 3). The studies showed moderate statistical heterogeneity ($I^2 = 50\%$ short term, 51% long term).

There were no significant differences between the groups with respect to short-term arm pain but there were statistically significant differences in favour of the standard fusion technique on long-term arm pain (3 studies; n = 174; Fig. 4). These results might be due to the fact that the standard deviations in one study might actually be standard errors [43]. We contacted the authors for clarification but they did not respond. When deleting this study from the analysis the result changed to nonsignificant. No heterogeneity was present.

Overall, the pooled effect estimates are small, varying from 0 to 2 mm on a VAS (Figs. 3 and 4) or a nonsignificant RD of 8–9% (data not pooled; see Supplementary Table 1).

3.4.1.3. Function. Eleven studies used the NDI as an outcome measure, of which 1 provided continuous data [53] and 3 dichotomised the score using a 10- or 15-point difference as the cutoff for a responder [33,48,65]. No other studies provided sufficient data. Pooling the results (3 studies; n = 190) showed a nonsignificant RD of 4% (−14% to 22%) (data not shown).

Overall, we found low-quality evidence (downgraded by limitations in design and publication bias) for no difference between the various fusion techniques on recovery. We found very low-quality evidence (downgraded by limitations in design, indirectness, and imprecision) for no differences between the various fusion techniques on pain and function (Table 1).

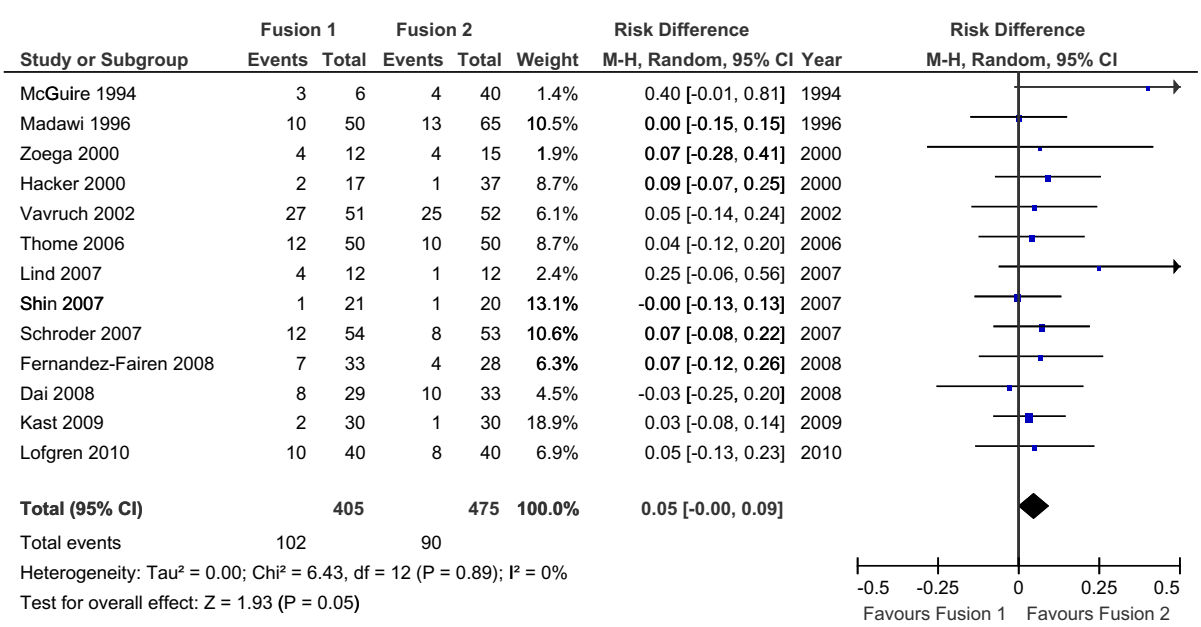

Fig. 2. Fusion vs fusion; clinical outcome at 24 months. Fusion 1, standard fusion technique; Fusion 2, experimental fusion technique.

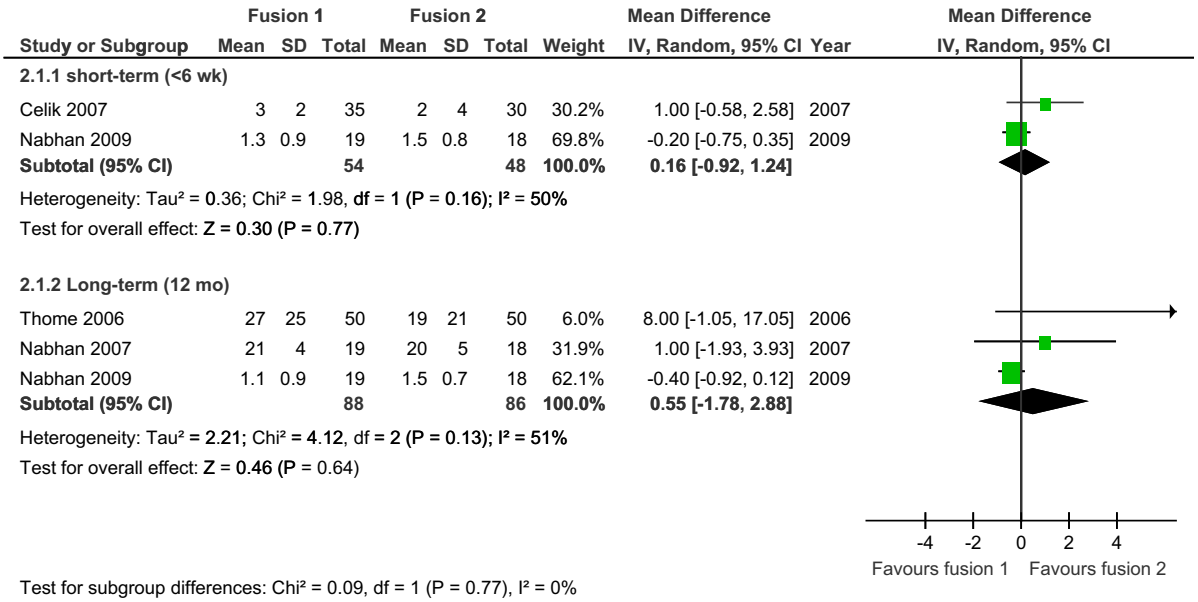

Fig. 3. Fusion vs fusion; neck pain. Fusion 1, standard fusion technique; Fusion 2, experimental fusion technique.

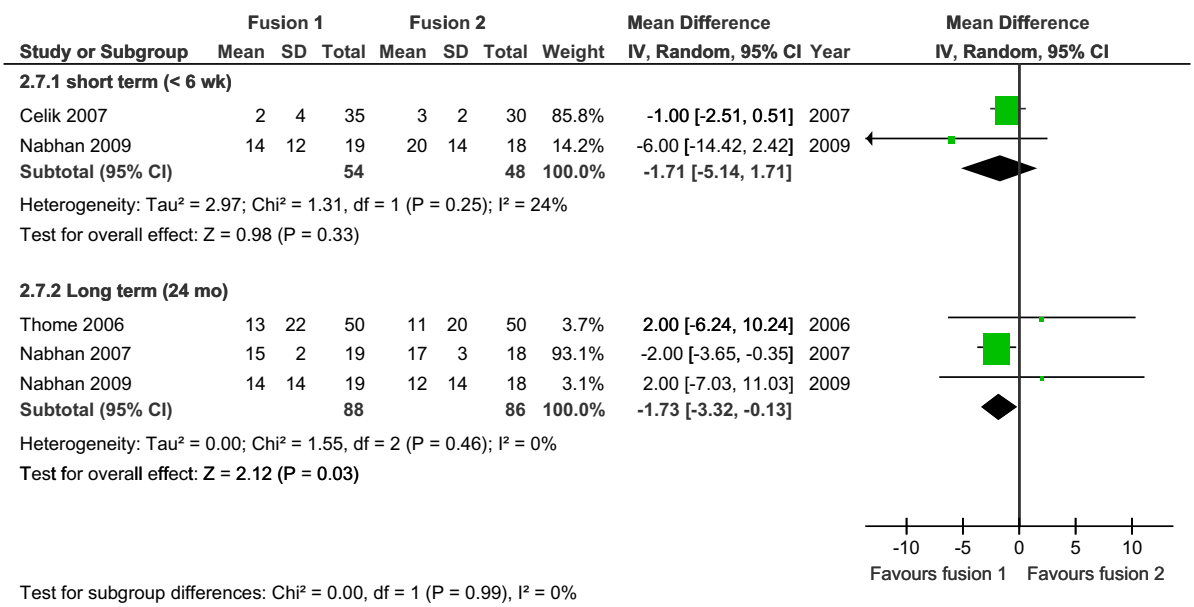

Fig. 4. Fusion vs fusion; arm pain. Fusion 1, standard fusion technique; Fusion 2, experimental fusion technique.

3.4.2. Surgical fusion methods compared with artificial (prosthetic) cervical disc surgery

3.4.2.1. Recovery. In 6 studies (n = 1640) recovery was measured using Odom's criteria, and all studies contributed to the statistical pooling (Fig. 5). Overall, 78.9% of patients recovered. There was a statistically significant difference in favour of prosthetic disc surgery (RD = 9%; 95% CI: 5–14). This difference was small and not considered clinically relevant. The only study with a low RoB [42] did not show a statistically significant difference. All studies reported a conflict of interest.

3.4.2.2. Pain. Four studies provided data on pain [27,43,66]. Pooled estimates of long-term neck (n = 275) and arm pain (n = 171) showed small but statistically significant differences of 1–3 mm on a VAS in favour of prosthetic disc surgery (data not shown).

3.4.2.3. Function. Four studies provided data on function (NDI) [25,27,42,66]. Moderate heterogeneity was present (I^2 = 55%) and pooled estimates of long-term function (n = 829) showed no difference between groups (data not shown).

3.4.2.4. Secondary surgery. Six studies (n = 1657) presented data on patients who needed revision surgery. The absolute risk of reoperation was 4.9%. The pooled summary estimate was not statistically significant: RD = 2% (−0% to 5%; Fig. 6).

For recovery we found moderate quality evidence (downgraded by limitations in design) for a statistically significant but clinically irrelevant effect of prosthetic disc surgery when compared with fusion. For pain we found low-quality evidence (downgraded by limitations in design, imprecision) that there were no differences between standard fusion and prosthetic disc surgery. Concerning

Table 1
Summary of findings.

Outcome	Quality of the evidence (GRADE [22])	Number of participants (studies)	Relative effect (95% CI)	Comparative absolute risk	
				Fusion 1	Fusion 2
Surgical fusion methods compared with other surgical fusion methods					
Recovery at long term (24 months)	Low[a,b]	880 (13 studies)	RD = 5 (−0; 9)	74.8%	81.1%
Neck pain (VAS or NRS)	Very low[a,c,d]				
Short term		102 (2 studies)	MD = 0.2 mm (−0.9; 1.2)		
Long term		174 (3 studies)	MD = 0.6 mm (−1.8; 2.9)		
Arm pain (VAS or NRS)	Very low[a,c,d]				
Short term		102 (2 studies)	MD = −1.7 mm (−5.1; 1.7)		
Long term		174 (3 studies)	MD = −1.7 mm (−3.3; 0.1)		
Function (NDI dichotomized)	Very low[a,c,d]	190 (3 studies)	RD = 4 (−14; 22)	37.2%	38.5%
Surgical fusion methods compared with artificial (prosthetic) cervical disc				Fusion	Prosthetic disc
Recovery at long term (24 months)	Moderate[a]	1640 (6 studies)	RD = 9 (4; 15)	75.4%	82.2%
Neck pain (VAS or NRS)	Low[a,d]				
Short term		41 (1 study)	MD = −8 mm (−5.2; −10.7)		
Long term		234 (3 studies)	MD = 1.4 mm (−0.1; 3.0)		
Arm pain (VAS or NRS)	Low[a,d]		MD = 3 mm (1.5; 4.6)MD = 1.6 (0.6; 2.7)		
Short term		41 (1 study)			
Long term		171 (3 studies)			
Function (NDI dichotomized)	Moderate[a]				
Long term		829 (4 studies)	SMD = 0.02 (−0.2; 0.2)		
Secondary surgery at long term (24 months)	Moderate[a]	1657 (6 studies)	RD = 2 (−0; 5)	6.0%	3.5%

Fusion 1, standard fusion technique; Fusion 2, experimental fusion technique; RD, risk difference; MD, mean difference; SMD, standardized mean difference; CI, confidence interval; NDI, neck disability index; NRS, numerical rating scale; VAS, visual analogue scale.
Positive values are in favour of fusion 2 or prosthetic disc.
[a] Downgraded by limitations in design.
[b] Downgraded because of publication bias.
[c] Downgraded by indirectness.
[d] Downgraded by imprecision.

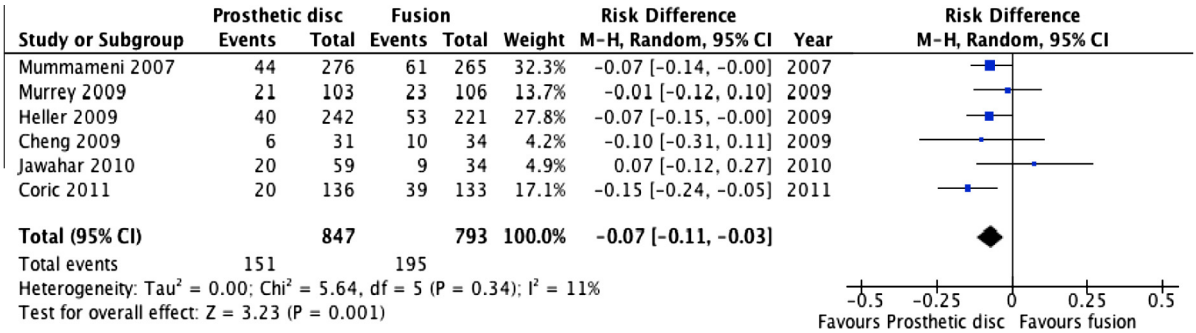

Fig. 5. Fusion vs prosthetic disc surgery; clinical outcome at 24 months.

reoperations and function we found moderate quality evidence (downgraded by limitations in design) for no significant differences between the groups.

4. Discussion

In this review of 39 RCTs we evaluated the effectiveness of various surgical techniques in patients with cervical disorders. We found no statistically significant and clinically relevant differences between fusion techniques, and statistically significant but clinically irrelevant differences in favour of prosthetic disc surgery compared with fusion. The authors of most studies of prosthetic disc surgery had a conflict of interest.

This review is the third of a set of 3 systematic reviews in which we focus on patient-relevant outcomes of surgery of the cervical spine [39,40]. The reviews all have the same a priori protocol and were split before the analysis. In one review (5 studies) we concluded that there was no clear benefit of surgery over conservative care, but all studies suffered from a high RoB. In another review (10 studies) we found no additional benefit of fusion techniques applied within an ACD procedure on pain, recovery, and return to work [40]. The current review shows apparently little difference in patient outcomes at both short- and long-term follow-up between various cervical surgical techniques. This is in line with 2 other recently published reviews, both evaluating cervical fusion in comparison with prosthetic disc surgery [4,17]. The choice for surgery should therefore be made with caution. When surgery is necessary the surgeon, health insurance company, and patient should base their choice of technique on surgical experience, preferences, or costs. To decide that there is real equivalence between

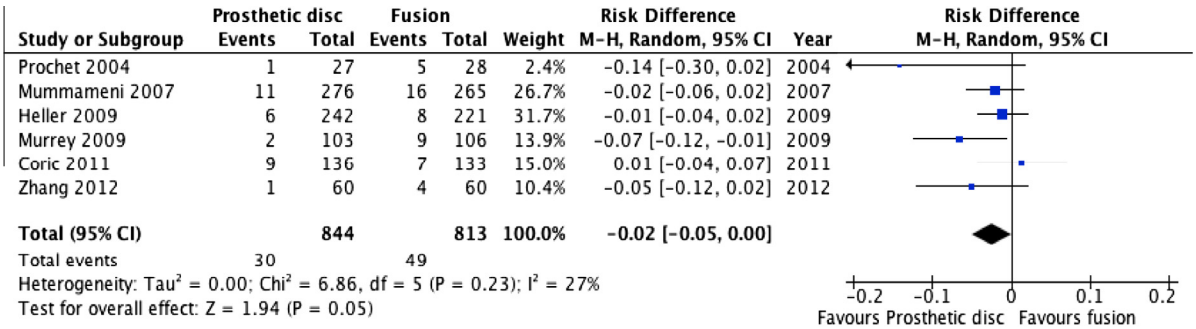

Fig. 6. Fusion vs prosthetic disc surgery; secondary surgery/reoperation.

cervical surgical interventions we need well-designed cost-effectiveness studies with sufficient power.

4.1. Methodological considerations

The methodological or reporting quality of most of the studies was poor; many criteria regarding the internal validity of the studies were unclear. Ten of the 39 studies did not present any point estimates or measures of variability and could not therefore contribute to the statistical pooling. We strongly recommend authors to follow the CONSORT guidelines for reporting quality to overcome these problems.

Very few studies reported blinding of patients, caregivers, and persons assessing the outcome measures. It remains unclear why patients (and when using patient-reported outcome measures also outcome measurement) were not blinded in these trials. When evaluating 2 surgical techniques it is feasible to blind the patients and caregivers (surgeons) in a way it does not lead to biased results. Surgeons, for instance, can be assigned to a single intervention, preferably the one they trust the most. Ensuring that a better blinding procedure is used is not only possible but also would greatly reduce the RoB in these studies.

4.2. Strengths and limitations

This review includes 39 RCTs and is therefore the largest systematic review of surgical intervention in cervical disorders. Previous reviews included 2–33 studies [4,17,26,30,39,40]. Nevertheless, we might have missed relevant unpublished trials, which are more likely to be small with nonsignificant results, leading to publication bias.

The inclusion of trials was difficult because relevant information on patient selection was frequently missing or unclear. This might be the explanation for the slight differences in the number of studies included among existing reviews. Just half of the studies stated they included, among others, patients with cervical radiculopathy. A recent review found that the classification of patients with cervical radiculopathy varies considerably [60]. We therefore chose not to regard them as a separate patient population, and subgroup analysis on recovery did not show different effect estimates. Also, some studies stated they included patients with myelopathy, but according to our clinical expert these patients are not considered to have distinct cervical myelopathy, which was considered an exclusion criterion.

We included studies that presented, among others, clinical outcomes, which might also be an explanation for the different numbers of studies included in the present review when compared with the others [4,17,30].

Screening the references of identified trials and systematic reviews may result in an over-representation of positive studies, as trials with a positive result are more likely to be referred to in other publications, leading to reference bias. In this review it was notable that there were multiple publications using the same dataset, with different numbers of subjects included without clear references to the other publications from the same dataset. Multiple publications influence the risk of publication bias, and when double publication cannot be detected it increases bias in the pooled effect estimates because patients are included in the analysis more than once. Different reviews appear to include different studies but actually include additional publications of the same trial, without referring to the issue of double publication.

It was apparent that some studies reported a mean follow-up time instead of an identical follow-up moment for all participants. This means that follow-up measurement has probably not been carried out according to a protocol and these analyses are therefore likely confounded.

Studies not published in English, Dutch, or German were not included in this review. It is not clear whether a language restriction is associated with bias and it is debatable what these studies would add to our findings [16].

Not all studies reported on the safety of the intervention. Most trials were also of low power and had short follow-up periods, making it difficult to evaluate surgical complications. Any conclusion on reoperations and other adverse events should therefore be considered with caution.

For future studies we recommend paying attention to patient selection and clearly stating the selection criteria used. Furthermore, we recommend that authors aim to blind patients and care providers, or at least assess their treatment preference. In the reporting of the study we strongly recommend future authors comply with the CONSORT statement.

Overall, we found no clinically relevant differences in any comparison or outcome measure. Nevertheless, the high RoB, low power of studies, a possible publication bias, and a conflict of interest of several authors might influence the results. For the time being any choice of surgical technique is up to the surgeon, patient, or health care provider.

Author contributions

Study concept and design: Verhagen A.P., Koes B. W., van Tulder M. W., Ostelo R., Peul W. C.

Analysis and interpretation of data: Verhagen A. P., van Middelkoop M., Rubinstein S. M. All had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Drafting of the manuscript: van Middelkoop M., Rubinstein S. M., Verhagen A. P.

Critical revision of the manuscript for important intellectual content: Peul W., Koes B. W., van Tulder M. W., Ostelo R., Verhagen A. P., Rubinstein S. M., Jacobs W. C. H.

Data analysis: Verhagen A. P., van Middelkoop M., Rubinstein S. M.

Obtained funding: Verhagen A. P., Koes B. W., van Tulder M. W., Ostelo R.

Study supervision: Koes B. W., van Tulder M. W., Verhagen A. P., Ostelo R., Peul W. C.

Financial disclosures

None reported.

Funding/support

This work was funded by the Dutch National Health Insurance Council.

Role of the sponsor

The funding organizations had no role in the design and conduct of the study; in the collection, analysis, and interpretation of the data; or in the preparation, review, or approval of the manuscript.

Additional contributions

We would like to thank Prof. E. J. Carraggee and Dr R. H. Bartels for their advice.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.pain.2013.07.022.

References

[1] Anderson PA, Sasso RC, Riew KD. Comparison of adverse events between the Bryan artificial cervical disc and anterior cervical arthrodesis. Spine (Phila Pa 1976) 2008;33:1305–12.
[2] Auerbach JD, Anakwenze OA, Milby AH, Lonner BS, Balderston RA. Segmental contribution toward total cervical range of motion: a comparison of cervical disc arthroplasty and fusion. Spine (Phila Pa 1976) 2011;36:E1593–9.
[3] Baskin DS, Ryan P, Sonntag V, Westmark R, Widmayer MA. A prospective, randomized, controlled cervical fusion study using recombinant human bone morphogenetic protein-2 with the CORNERSTONE-SR allograft ring and the ATLANTIS anterior cervical plate. Spine (Phila Pa 1976) 2003;28:1219–24 [discussion 1225].
[4] Boselie TFM, Willems PC, van Mameren H, de Bie R, Benzel EC, van Santbrink H. Arthroplasty versus fusion in single-level cervical degenerative disc disease. Cochrane Database Syst Rev 2012(9):CD009173.
[5] Burkus JK, Haid Jr RW, Traynelis VC, Mummaneni PV. Long-term clinical and radiographic outcomes of cervical disc replacement with the Prestige disc: results from a prospective randomized controlled clinical trial. J Neurosurg Spine 2010;13:308–18.
[6] Celik SE, Kara A, Celik S. A comparison of changes over time in cervical foraminal height after tricortical iliac graft or polyetheretherketone cage placement following anterior discectomy. J Neurosurg Spine 2007;6:10–6.
[7] Cheng L, Nie L, Li M, Huo Y, Pan X. Superiority of the Bryan® disc prosthesis for cervical myelopathy: a randomized study with 3-year followup. Clin Orthop Relat Res 2011;469:3408–14.
[8] Cheng L, Nie L, Zhang L, Hou Y. Fusion versus Bryan Cervical Disc in two-level cervical disc disease: a prospective, randomised study. Int Orthop 2009;33:1347–51.
[9] Cho DY, Lee WY, Sheu PC, Chen CC. Cage containing a biphasic calcium phosphate ceramic (Triosite) for the treatment of cervical spondylosis. Surg Neurol 2005;63:497–503 [discussion 503–94].
[10] Cho DY, Lee WY, Sheu PC. Treatment of multilevel cervical fusion with cages. Surg Neurol 2004;62:378–85 [discussion 385–76].
[11] Coric D, Finger F, Boltes P. Prospective randomized controlled study of the Bryan Cervical Disc: early clinical results from a single investigational site. J Neurosurg Spine 2006;4:31–5.
[12] Coric D, Nunley PD, Guyer RD, Musante D, Carmody CN, Gordon CR, Lauryssen C, Ohnmeiss DD, Boltes MO. Prospective, randomized, multicenter study of cervical arthroplasty: 269 patients from the Kineflex|C artificial disc investigational device exemption study with a minimum 2-year follow-up: clinical article. J Neurosurg Spine 2011;15:348–58.
[13] Cote P, Cassidy JD, Carroll L. The Saskatchewan Health and Back Pain Survey. The prevalence of neck pain and related disability in Saskatchewan adults. Spine (Phila Pa 1976) 1998;23:1689–98.
[14] Dai LY, Jiang LS. Anterior cervical fusion with interbody cage containing beta-tricalcium phosphate augmented with plate fixation: a prospective randomized study with 2-year follow-up. Eur Spine J 2008;17:698–705.
[15] Delamarter RB, Murrey D, Janssen ME, Goldstein JA, Zigler J, Tay BKB, Darden B. Results at 24 months from the prospective, randomized, multicenter Investigational Device Exemption trial of ProDisc-C versus anterior cervical discectomy and fusion with 4-year follow-up and continued access patients. SAS J 2010;4:122–8.
[16] Egger M, Smith GD. Bias in location and selection of studies. BMJ 1998;316:61–6.
[17] Fallah A, Akl EA, Ebrahim S, Ibrahim GM, Mansouri A, Foote CJ, Zhang Y, Fehlings MG. Anterior cervical discectomy with arthroplasty versus arthrodesis for single-level cervical spondylosis: a systematic review and meta-analysis. PLoS One 2012;7:e43407.
[18] Feiz-Erfan I, Harrigan M, Sonntag VK, Harrington TR. Effect of autologous platelet gel on early and late graft fusion in anterior cervical spine surgery. J Neurosurg Spine 2007;7:496–502.
[19] Fernandez-Fairen M, Sala P, Dufoo Jr M, Ballester J, Murcia A, Merzthal L. Anterior cervical fusion with tantalum implant: a prospective randomized controlled study. Spine (Phila Pa 1976) 2008;33:465–72.
[20] Furlan AD, Pennick V, Bombardier C, van Tulder M. 2009 updated method guidelines for systematic reviews in the Cochrane Back Review Group. Spine (Phila Pa 1976) 2009;34:1929–41.
[21] Grob D, Peyer JV, Dvorak J. The use of plate fixation in anterior surgery of the degenerative cervical spine: a comparative prospective clinical study. Eur Spine J 2001;10:408–13.
[22] Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, Schünemann HJ, GRADE Working Group. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ 2008; 336:924–6.
[23] Hacker RJ. A randomized prospective study of an anterior cervical interbody fusion device with a minimum of 2 years of follow-up results. J Neurosurg 2000;93:222–6.
[24] Hacker RJ. Cervical disc arthroplasty: a controlled randomized prospective study with intermediate follow-up results. Invited submission from the joint section meeting on disorders of the spine and peripheral nerves, March 2005. J Neurosurg Spine 2005;3:424–8.
[25] Heller JG, Sasso RC, Papadopoulos SM, Anderson PA, Fessler RG, Hacker RJ, Coric D, Cauthen JC, Riew DK. Comparison of BRYAN cervical disc arthroplasty with anterior cervical decompression and fusion: clinical and radiographic results of a randomized, controlled, clinical trial. Spine (Phila Pa 1976) 2009;34:101–7.
[26] Jacobs W, Willems PC, van Limbeek J, Bartels R, Pavlov P, Anderson PG, Oner C. Single or double-level anterior interbody fusion techniques for cervical degenerative disc disease. Cochrane Database Syst Rev 2011;1:5–8.
[27] Jawahar A, Cavanaugh DA, Kerr EJ, Birdsong EM, Nunley PD. Total disc arthroplasty does not affect the incidence of adjacent segment degeneration in cervical spine: results of 93 patients in three prospective randomized clinical trials. Spine J 2010;10:1043–8.
[28] Kast E, Derakhshani S, Bothmann M, Oberle J. Subsidence after anterior cervical inter-body fusion. A randomized prospective clinical trial. Neurosurg Rev 2009;32:207–14.
[29] Kelly MP, Mok JM, Frisch RF, Tay BK. Adjacent segment motion after anterior cervical discectomy and fusion versus ProDisc-C cervical total disk arthroplasty: analysis from a randomized, controlled trial. Spine (Phila Pa 1976) 2011;36:1171–9.
[30] van Limbeek J, Jacobs WC, Anderson PG, Pavlov PW. A systematic literature review to identify the best method for a single level anterior cervical interbody fusion. Eur Spine J 2000;9:129–36.
[31] Lind BI, Zoega B, Rosen H. Autograft versus interbody fusion cage without plate fixation in the cervical spine: a randomized clinical study using radiostereometry. Eur Spine J 2007;16:1251–6.
[32] Linton SJ, Hellsing AL, Hallden K. A population-based study of spinal pain among 35–45-year-old individuals. Prevalence, sick leave, and health care use. Spine (Phila Pa 1976) 1998;23:1457–63.
[33] Lofgren HL, Engquist M, Hoffmann P, Sigstedt B, Vavruch L. Clinical and radiological evaluation of Trabecular Metal and the Smith-Robinson technique in anterior cervical fusion for degenerative disease: a prospective, randomized, controlled study with 2-year follow-up. Eur Spine J 2010;19:464–73.
[34] Madawi AA, Powell M, Crockard HA. Biocompatible osteoconductive polymer versus iliac graft. A prospective comparative study for the evaluation of fusion pattern after anterior cervical discectomy. Spine (Phila Pa 1976) 1996;21:2123–9 [discussion 2129–30].
[35] Makela M, Heliovaara M, Sievers K, Impivaara O, Knekt P, Aromaa A. Prevalence, determinants, and consequences of chronic neck pain in Finland. Am J Epidemiol 1991;134:1356–67.
[36] McConnell JR, Freeman BJ, Debnath UK, Grevitt MP, Prince HG, Webb JK. A prospective randomized comparison of coralline hydroxyapatite with autograft in cervical interbody fusion. Spine (Phila Pa 1976) 2003;28: 317–23.
[37] McGuire RA, St John K. Comparison of anterior cervical fusions using autogenous bone graft obtained from the cervical vertebrae to the modified Smith-Robinson technique. J Spinal Disord 1994;7:499–503.
[38] Meier U, Kemmesies D. Experiences with six different intervertebral disc spacers for spondylodesis of the cervical spine. Orthopade 2004;33: 1290–9.

[39] van Middelkoop M, Rubinstein SM, Ostelo R, van Tulder MW, Peul W, Koes BW, Verhagen AP. Surgery versus conservative care for neck pain: a systematic review. Eur Spine J 2013;22:87–95.
[40] van Middelkoop M, Rubinstein SM, Ostelo R, van Tulder MW, Peul W, Koes BW, Verhagen AP. No additional value of fusion techniques on anterior discectomy for neck pain: a systematic review. PAIN® 2012;153:2167–73.
[41] Mummaneni PV, Burkus JK, Haid RW, Traynelis VC, Zdeblick TA. Clinical and radiographic analysis of cervical disc arthroplasty compared with allograft fusion: a randomized controlled clinical trial. J Neurosurg Spine 2007;6:198–209.
[42] Murrey D, Janssen M, Delamarter R, Goldstein J, Zigler J, Tay B, Darden B. Results of the prospective, randomized, controlled multicenter Food and Drug Administration investigational device exemption study of the ProDisc-C total disc replacement versus anterior discectomy and fusion for the treatment of 1-level symptomatic cervical disc disease. Spine J 2009;9:275–86.
[43] Nabhan A, Ahlhelm F, Shariat K, Pitzen T, Steimer O, Steudel WI, Pape D. The ProDisc-C prosthesis: clinical and radiological experience 1 year after surgery. Spine (Phila Pa 1976) 2007;32:1935–41.
[44] Nabhan A, Ishak B, Steimer O, Zimmer A, Pitzen T, Steudel WI, Pape D. Comparison of bioresorbable and titanium plates in cervical spinal fusion: early radiologic and clinical results. J Spinal Disord Tech 2009;22:155–61.
[45] Nabhan A, Ishak B, Steudel WI, Ramadhan S, Steimer O. Assessment of adjacent-segment mobility after cervical disc replacement versus fusion: RCT with 1 year's results. Eur Spine J 2011;20:934–41.
[46] Nabhan A, Pape D, Pitzen T, Steudel WI, Bachelier F, Jung J, Ahlhelm F. Radiographic analysis of fusion progression following one-level cervical fusion with or without plate fixation. Zentralbl Neurochir 2007;68:133–8.
[47] Nunley PD, Jawahar A, Kerr 3rd EJ, Cavanaugh DA, Howard C, Brandao SM. Choice of plate may affect outcomes for single versus multilevel ACDF: results of a prospective randomized single-blind trial. Spine J 2009;9:121–7.
[48] Pitzen TR, Chrobok J, Stulik J, Ruffing S, Drumm J, Sova L, Kucera R, Vyskocil T, Steudel WI. Implant complications, fusion, loss of lordosis, and outcome after anterior cervical plating with dynamic or rigid plates: two-year results of a multi-centric, randomized, controlled study. Spine 2009;34:641–6.
[49] Porchet F, Metcalf NH. Clinical outcomes with the Prestige II cervical disc: preliminary results from a prospective randomized clinical trial. Neurosurg Focus 2004;17:E6.
[50] Rajala U, Keinanen-Kiukaanniemi S, Uusimaki A, Kivela SL. Musculoskeletal pains and depression in a middle-aged Finnish population. PAIN® 1995;61:451–7.
[51] Ryu SI, Mitchell M, Kim DH. A prospective randomized study comparing a cervical carbon fiber cage to the Smith-Robinson technique with allograft and plating: up to 24 months follow-up. Eur Spine J 2006;15:157–64.
[52] Sasso RC, Anderson PA, Riew KD, Heller JG. Results of cervical arthroplasty compared with anterior discectomy and fusion: four-year clinical outcomes in a prospective, randomized controlled trial. J Bone Joint Surg Am 2011;93:1684–92.
[53] Sasso RC, Anderson PA, Riew KD, Heller JG. Results of cervical arthroplasty compared with anterior discectomy and fusion: four-year clinical outcomes in a prospective, randomized controlled trial. Orthopedics 2011;34:889.
[54] Sasso RC, Best NM. Cervical kinematics after fusion and Bryan disc arthroplasty. J Spinal Disord Tech 2008;21:19–22.
[55] Schroder J, Grosse-Dresselhaus F, Schul C, Wassmann H. PMMA versus titanium cage after anterior cervical discectomy—a prospective randomized trial. Zentralbl Neurochir 2007;68:2–7.
[56] Shin SH, Lee WJ, Eun JP, Choi HY, Lee JC. Clinical and radiologic assessment for anterior cervical interbody fusion with synthetic cages. J Korean Neurosurg Soc 2007;41:105–10.
[57] Siddiqui AA, Jackowski A. Cage versus tricortical graft for cervical interbody fusion. A prospective randomised study. J Bone Joint Surg Br 2003;85:1019–25.
[58] Steinmetz MP, Patel R, Traynelis V, Resnick DK, Anderson PA. Cervical disc arthroplasty compared with fusion in a workers' compensation population. Neurosurgery 2008;63:741–7 [discussion 747].
[59] Thome C, Leheta O, Krauss JK, Zevgaridis D. A prospective randomized comparison of rectangular titanium cage fusion and iliac crest autograft fusion in patients undergoing anterior cervical discectomy. J Neurosurg Spine 2006;4:1–9.
[60] Thoomes EJ, Scholten-Peeters GG, de Boer AJ, Olsthoorn RA, Verkerk K, Lin C, Verhagen AP. Lack of uniform diagnostic criteria for cervical radiculopathy in conservative intervention studies: a systematic review. Eur Spine J. 2012;21:1459–70.
[61] van Tulder MW, Suttorp M, Morton S, Bouter LM, Shekelle P. Empirical evidence of an association between internal validity and effect size in randomized controlled trials of low-back pain. Spine 2009;34:1685–92.
[62] Vavruch L, Hedlund R, Javid D, Leszniewski W, Shalabi A. A prospective randomized comparison between the Cloward procedure and a carbon fiber cage in the cervical spine: a clinical and radiologic study.. Spine (Phila Pa 1976) 2002;27:1694–701.
[63] Villavicencio AT, Babuska JM, Ashton A, Busch E, Roeca C, Lee Nelson E, Mason A, Burneikiene S. Prospective, randomized, double blind, clinical study evaluating the correlation of clinical outcomes and cervical sagittal alignment. Neurosurgery 2011;68:1309–16.
[64] Vos T, Flaxman A, Naghavl M, Lozano R, Michaud C, Ezzati M, et al. Years lived with disability (YLDs) for 1160 sequelae of 289 diseases and injuries 1990–2010: a systematic analysis for the Global Burden of Disease Study 2010. Lancet 2012;380:2163–96.
[65] Wigfield C, Robertson J, Gill S, Nelson R. Clinical experience with porous tantalum cervical interbody implants in a prospective randomized controlled trial. Br J Neurosurg 2003;17:418–25.
[66] Zhang X, Zhang X, Chen C, Zhang Y, Wang Z, Wang B, Yan W, Li M, Yuan W, Wang Y. Randomized, controlled, multicenter, clinical trial comparing BRYAN cervical disc arthroplasty with anterior cervical decompression and fusion in China. Spine (Phila Pa 1976) 2012;37:433–8.
[67] Zoega B, Karrholm J, Lind B. One-level cervical spine fusion. A randomized study, with or without plate fixation, using radiostereometry in 27 patients. Acta Orthop Scand 1998;69:363–8.
[68] Zoega B, Karrholm J, Lind B. Plate fixation adds stability to two-level anterior fusion in the cervical spine: a randomized study using radiostereometry. Eur Spine J 1998;7:302–7.
[69] Zoega B, Rosen H, Lind B. Anterior cervical discectomy and fusion with or without plate fixation. A prospective and randomized study. Euro-Orthopedics 2000;28:39–51.